

Quiz (Carolina Reaper)

- Q1.** A shipping company has a cost function $C(x) = 2x^2 + 5x + 3$ to transport x crates. Factor the function.
- (A) $C(x) = (2x + 1)(x + 3)$
(B) $C(x) = (2x + 3)(x + 1)$
(C) $C(x) = (x + 1)(2x + 3)$
(D) $C(x) = (x + 3)(2x + 1)$
(E) $C(x) = (2x + 3)(x - 1)$
- Q2.** The time t it takes for a package to be delivered is given by $t(x) = x^2 + 4x + 4$. Factor $t(x)$.
- (A) $t(x) = (x + 2)(x - 2)$
(B) $t(x) = (x + 2)(x + 2)$
(C) $t(x) = (x - 2)(x - 2)$
(D) $t(x) = (x + 4)(x + 1)$
(E) $t(x) = (x - 4)(x - 1)$
- Q3.** A traffic flow model is given by $F(x) = 3x^2 + 2x - 1$. Factor $F(x)$ if possible.
- (A) $F(x) = (3x - 1)(x + 1)$
(B) $F(x) = (3x + 1)(x - 1)$
(C) $F(x) = (x + 1)(3x - 1)$
(D) Not factorable
(E) $F(x) = (x - 1)(3x + 1)$
- Q4.** A logistics company has a profit function $P(x) = 2x^2 - 5x - 3$. Factor $P(x)$ if possible.
- (A) $P(x) = (2x + 1)(x - 3)$
(B) $P(x) = (2x - 1)(x + 3)$
(C) $P(x) = (x + 1)(2x - 3)$
(D) Not factorable
(E) $P(x) = (x - 1)(2x + 3)$
- Q5.** The cost of shipping x packages is given by $C(x) = x^2 + 2x + 1$. Factor $C(x)$.
- (A) $C(x) = (x + 1)(x - 1)$
(B) $C(x) = (x + 1)(x + 1)$
(C) $C(x) = (x - 1)(x - 1)$
(D) $C(x) = (x + 2)(x + 1)$
(E) $C(x) = (x - 2)(x - 1)$
- Q6.** The time t it takes for a truck to travel x miles is given by $t(x) = x^2 - 4x + 4$. Factor $t(x)$.
- (A) $t(x) = (x + 2)(x - 2)$
(B) $t(x) = (x - 2)(x - 2)$
(C) $t(x) = (x + 2)(x + 2)$
(D) $t(x) = (x + 4)(x + 1)$
(E) $t(x) = (x - 4)(x - 1)$
- Q7.** A delivery company has a function $D(x) = x^2 + 5x + 6$ to model the number of deliveries. Factor $D(x)$.
- (A) $D(x) = (x + 2)(x + 3)$
(B) $D(x) = (x + 1)(x + 6)$
(C) $D(x) = (x - 2)(x + 3)$
(D) $D(x) = (x + 2)(x - 3)$
(E) $D(x) = (x - 1)(x + 6)$
- Q8.** The cost of fuel for a truck is given by $F(x) = 2x^2 + 7x + 3$. Factor $F(x)$ if possible.
- (A) $F(x) = (2x + 1)(x + 3)$
(B) $F(x) = (2x + 3)(x + 1)$
(C) $F(x) = (x + 3)(2x + 1)$
(D) Not factorable
(E) $F(x) = (x + 1)(2x + 3)$

Open Ended Questions (FRQ)

- Q9.** A company has a cost function $C(x) = ax^2 + bx + c$ to transport x crates. If $C(1) = 10$, $C(2) = 25$, and $C(3) = 42$, find the factored form of $C(x)$.
- Q10.** A logistics company has a profit function $P(x) = 3x^2 - 2x - 1$. Factor $P(x)$ if possible, and explain your reasoning.

Chef's Tips

- Read each question carefully before answering.
- Show all work and explain your reasoning for FRQ questions.
- Use the given formulas and equations to solve problems.